Zero-Leakage 4-Way Chronological Dataset Partitioning Architecture

Strict Causal Timeline Isolation: Eliminating Lookahead Bias, Graph Leakage, and Conformal Contamination

[Pillar 1] Causal Graph Flow	[Pillar 2] Quarantined SMOTE	[Pillar 3] Finite Exchangeability	[Pillar 4] Out-of-Time Serving
Past only: $\tau_{\text{past}} \le t$	Confined strictly to $\mathcal{D}_{\text{train}}$	$\mathcal{D}_{\text{cal}} \cap \mathcal{D}_{\text{val}} = \varnothing$ (Disjoint)	Unseen future stream evaluation

Track A: Discrete UTXO Graph Snapshots (Elliptic-v1 & Elliptic-v2: 49 Timesteps, 2-Week Windows):

Time Arrow →



Track B: Continuous High-Velocity Streams (PaySim, SAML-D, IBM-AMLSim, Ethereum-Phish):



1. Training Split ($\mathcal{D}_{\text{train}}$: 60%)
Model Weights & Topology Optimization

- **Target: Parameter Learning**
  Optimizes neural weights Θ for STGB.
- **Scope: Past-Only Subgraphs**
  Strict $\mathcal{G}_{\le t_{\text{train}}}$; zero future edges.
- **Quarantine: Latent SMOTE**
  Synthetic fraud nodes confined here.
- **Slices: 60% Chrono Horizon**
  Steps 1–30 | Stream 0.00 to 0.60 T.

2. Validation Split ($\mathcal{D}_{\text{val}}$: 10%)
Checkpointing & Bayes Threshold

- **Tuning: Bayes Threshold τ^***
  N=50 Bayesian hyperparameter trials.
- **Decision: Macro F1 Optimization**
  Calibrates optimal cutoff τ^*.
- **Lock: Zero Gradient Feedback**
  Model weights Θ frozen during eval.
- **Slices: 10% Chrono Horizon**
  Steps 31–34 | Stream 0.60 to 0.70 T.

3. Calibration Split ($\mathcal{D}_{\text{cal}}$: 10%)
Class-Conditional Risk Quantiles

- **CRC: Conformal Risk Control**
  Computes class quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$.
- **Coverage: Finite Exchangeability**
  Guarantees risk coverage $1 - \alpha \ge 99\%$.
- **Isolation: Disjoint Holdout**
  Zero overlap with train or val sets.
- **Slices: 10% Chrono Horizon**
  Steps 35–38 | Stream 0.70 to 0.80 T.

4. Forward Test Split ($\mathcal{D}_{\text{test}}$: 20%)
Out-of-Time Forward Deployment

- **Benchmark: Out-of-Sample**
  Final reporting across 5 locked seeds.
- **Production: Streaming SLA**
  Simulates live deployment on unseen tx.
- **Zero Peeking: Pure Forward Eval**
  Strict one-pass scoring without updates.
- **Slices: 20% Chrono Horizon**
  Steps 39–49 | Stream 0.80 to 1.00 T.

METHODOLOGICAL RIGOR: Zero-Leakage Guarantee Across All 14 Financial Transaction Networks

Strict Temporal Causality: Past Information Only ($\tau_{\text{past}} \le t$) | Disjoint Calibration Set $\mathcal{D}_{\text{cal}}$ Guarantees Valid Conformal Risk Coverage ($1 - \alpha \ge 99.0\%$)